

LIMESTONE AND KARST LANDFORMS / INDIA CAVES-MEGHALAYAN AGE

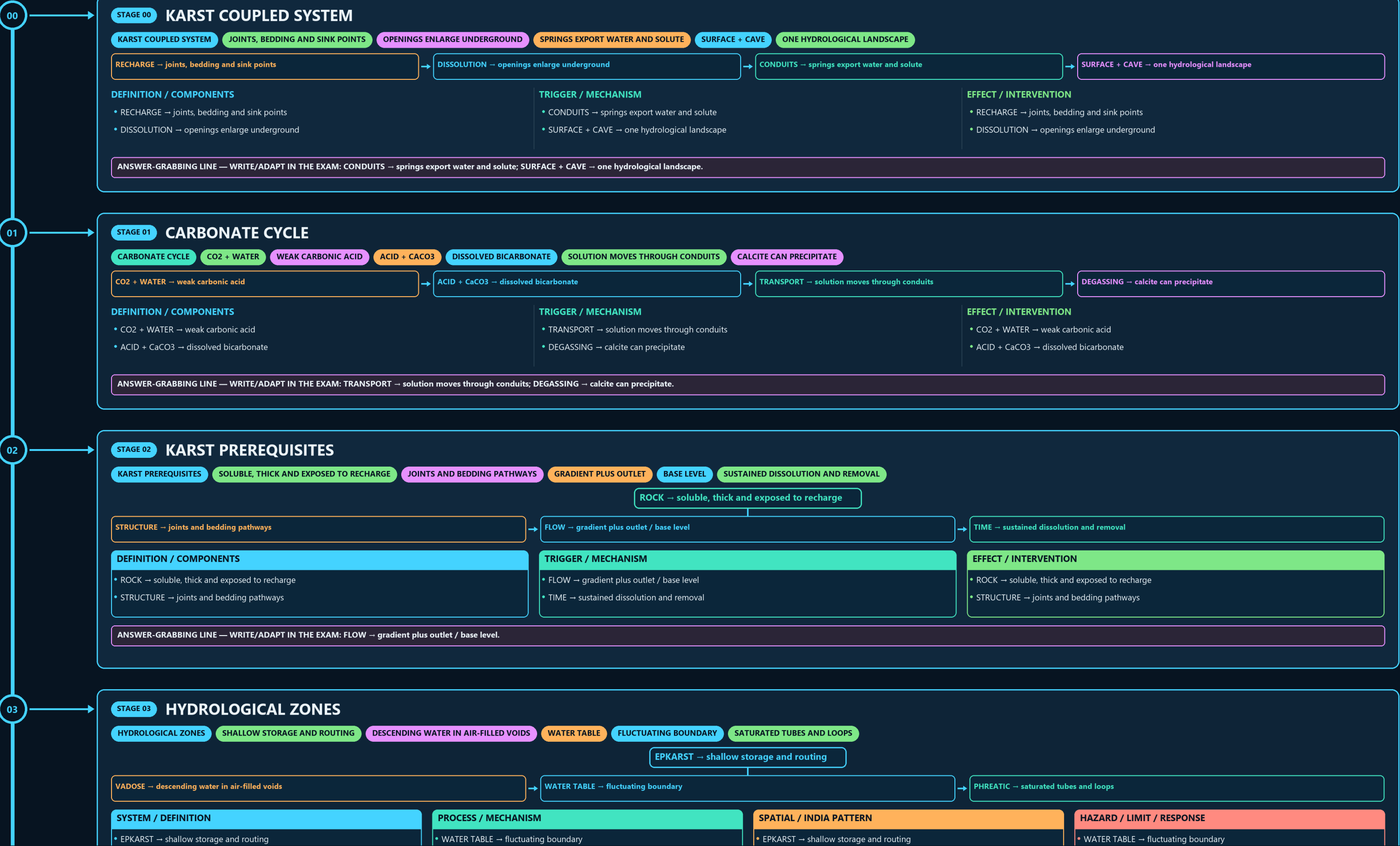
KARST COUPLED SYSTEM → CARBONATE CYCLE → KARST PREREQUISITES → HYDROLOGICAL ZONES → SURFACE-KARST LADDER → CAVE-DEVELOPMENT SEQUENCE

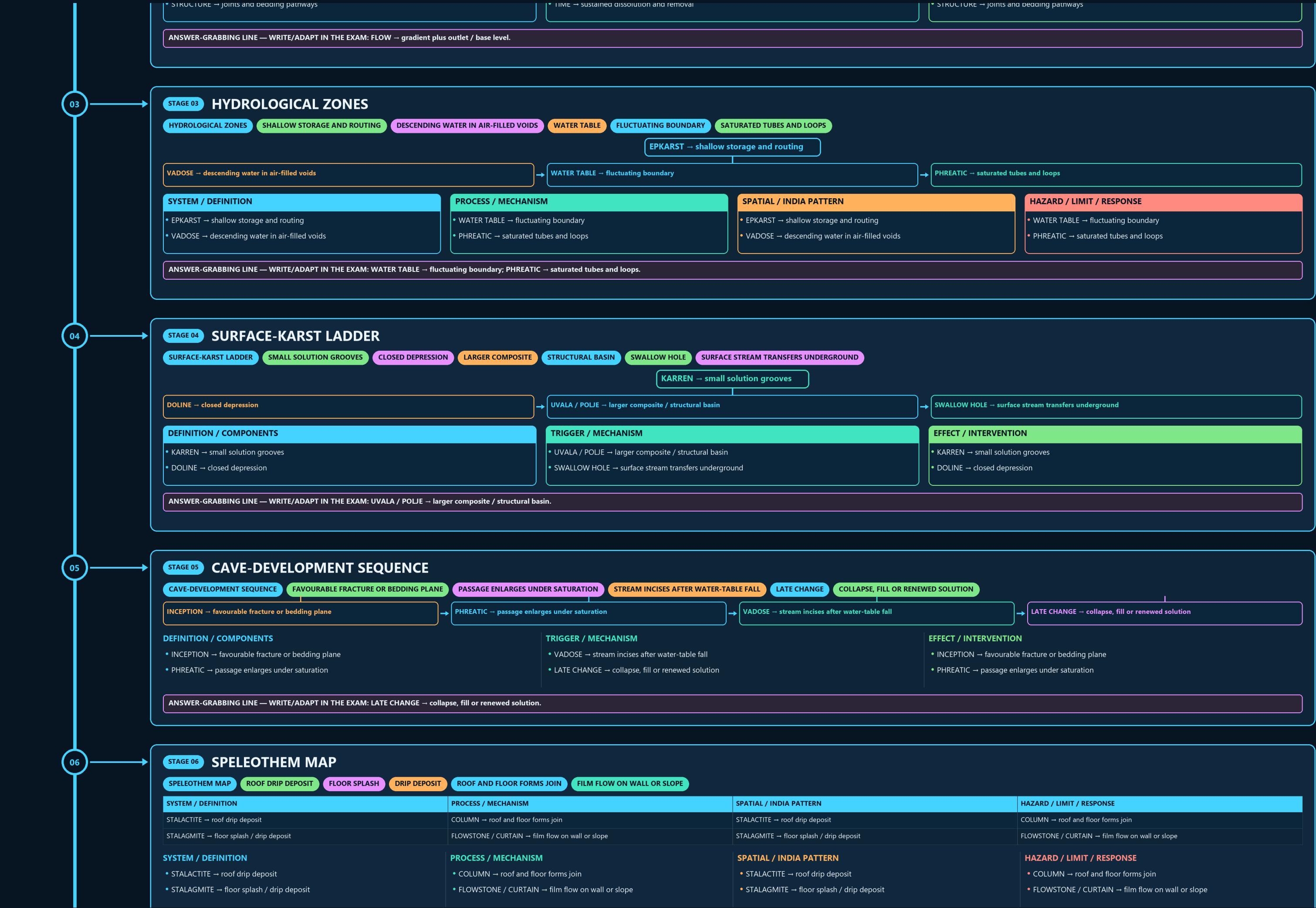
Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g3 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles





ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LATE CHANGE → collapse, fill or renewed solution.

06

STAGE 06 SPELEOTHEM MAP

SPELEOTHEM MAP ROOF DRIP DEPOSIT FLOOR SPLASH DRIP DEPOSIT ROOF AND FLOOR FORMS JOIN FILM FLOW ON WALL OR SLOPE

SYSTEM / DEFINITION	PROCESS / MECHANISM	SPATIAL / INDIA PATTERN	HAZARD / LIMIT / RESPONSE
STALACTITE → roof drip deposit	COLUMN → roof and floor forms join	STALACTITE → roof drip deposit	COLUMN → roof and floor forms join
STALAGMITE → floor splash / drip deposit	FLOWSTONE / CURTAIN → film flow on wall or slope	STALAGMITE → floor splash / drip deposit	FLOWSTONE / CURTAIN → film flow on wall or slope
SYSTEM / DEFINITION - STALACTITE → roof drip deposit - STALAGMITE → floor splash / drip deposit	PROCESS / MECHANISM - COLUMN → roof and floor forms join - FLOWSTONE / CURTAIN → film flow on wall or slope	SPATIAL / INDIA PATTERN - STALACTITE → roof drip deposit - STALAGMITE → floor splash / drip deposit	HAZARD / LIMIT / RESPONSE - COLUMN → roof and floor forms join - FLOWSTONE / CURTAIN → film flow on wall or slope

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FLOWSTONE / CURTAIN → film flow on wall or slope.

07

STAGE 07 INDIA CAVE FIREWALL

INDIA CAVE FIREWALL MEGHALAYA LIMESTONE NATURAL CAVES IN ANDHRA PRADESH HUMAN ROCK-CUT ARCHITECTURE LOCATION DOES NOT REPLACE GENETIC CLASSIFICATION

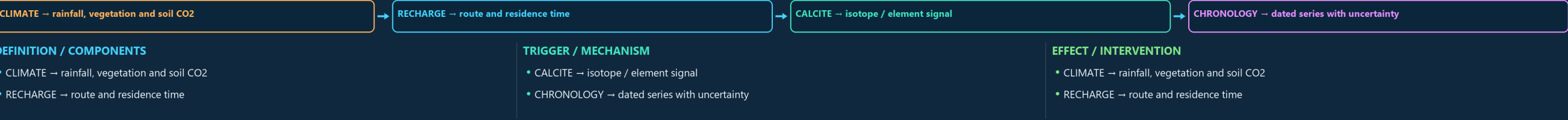
SYSTEM / DEFINITION	PROCESS / MECHANISM	SPATIAL / INDIA PATTERN	HAZARD / LIMIT / RESPONSE
MAWMLUH / MAWSMAI / SIJU → Meghalaya limestone	AJANTA / ELLORA → human rock-cut architecture	MAWMLUH / MAWSMAI / SIJU → Meghalaya limestone	AJANTA / ELLORA → human rock-cut architecture
BORRA / BELUM → natural caves in Andhra Pradesh	RULE → location does not replace genetic classification	BORRA / BELUM → natural caves in Andhra Pradesh	RULE → location does not replace genetic classification
SYSTEM / DEFINITION - MAWMLUH / MAWSMAI / SIJU → Meghalaya limestone - BORRA / BELUM → natural caves in Andhra Pradesh	PROCESS / MECHANISM - AJANTA / ELLORA → human rock-cut architecture - RULE → location does not replace genetic classification	SPATIAL / INDIA PATTERN - MAWMLUH / MAWSMAI / SIJU → Meghalaya limestone - BORRA / BELUM → natural caves in Andhra Pradesh	HAZARD / LIMIT / RESPONSE - AJANTA / ELLORA → human rock-cut architecture - RULE → location does not replace genetic classification

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → location does not replace genetic classification.

08

STAGE 08 PROXY PIPELINE

PROXY PIPELINE RAINFALL, VEGETATION AND SOIL CO2 ROUTE AND RESIDENCE TIME ELEMENT SIGNAL DATED SERIES WITH UNCERTAINTY

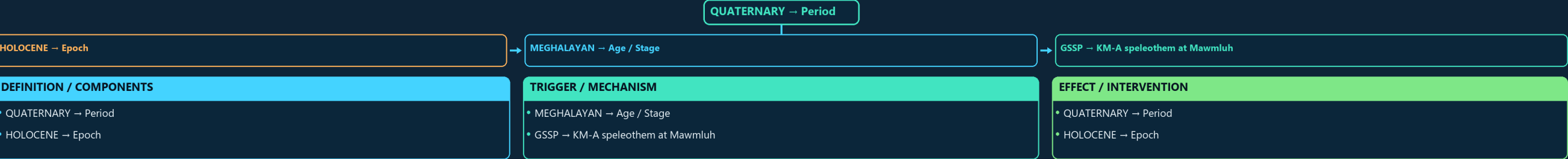


ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CALCITE → isotope / element signal; CHRONOLOGY → dated series with uncertainty.

09

STAGE 09 MEGHALAYAN HIERARCHY

MEGHALAYAN HIERARCHY KM-A SPELEOTHEM AT MAWMLUH QUATERNARY → PERIOD HOLOCENE → EPOCH



ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MEGHALAYAN → Age / Stage; GSSP → KM-A speleothem at Mawmluh.

09

STAGE 09

MEGHALAYAN HIERARCHY

MEGHALAYAN HIERARCHY

KM-A SPELEOTHEM AT MAWMLUH

QUATERNARY → PERIOD

HOLOCENE → EPOCH

QUATERNARY → Period

HOLOCENE → Epoch

MEGHALAYAN → Age / Stage

GSSP → KM-A speleothem at Mawmluh

DEFINITION / COMPONENTS

- QUATERNARY → Period
- HOLOCENE → Epoch

TRIGGER / MECHANISM

- MEGHALAYAN → Age / Stage
- GSSP → KM-A speleothem at Mawmluh

EFFECT / INTERVENTION

- QUATERNARY → Period
- HOLOCENE → Epoch

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MEGHALAYAN → Age / Stage; GSSP → KM-A speleothem at Mawmluh.

10

STAGE 10

STATUS AND NARRATIVE CAUTION

STATUS AND NARRATIVE CAUTION

4.2 KA EVENT

CORRELATION GUIDE

GSSP POINT

PHYSICAL LOWER BOUNDARY

NAMED FORMAL INTERVAL

NOT A RATIFIED EPOCH

DEFINITION / COMPONENTS

- 4.2 ka EVENT → correlation guide
- GSSP POINT → physical lower boundary

TRIGGER / MECHANISM

- MEGHALAYAN → named formal interval
- ANTHROPOCENE → not a ratified Epoch

EFFECT / INTERVENTION

- 4.2 ka EVENT → correlation guide
- GSSP POINT → physical lower boundary

DEFINITION / COMPONENTS

- 4.2 ka EVENT → correlation guide
- GSSP POINT → physical lower boundary

TRIGGER / MECHANISM

- MEGHALAYAN → named formal interval
- ANTHROPOCENE → not a ratified Epoch

EFFECT / INTERVENTION

- 4.2 ka EVENT → correlation guide
- GSSP POINT → physical lower boundary

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MEGHALAYAN → named formal interval; ANTHROPOCENE → not a ratified Epoch.

11

STAGE 11

KARST CONSERVATION SPINE

KARST CONSERVATION SPINE

RECHARGE, PASSAGES, SPRINGS AND VALUES

IRREVERSIBLE QUARRY AND POLLUTION PATHWAYS

VISITATION, WASTE AND WATER USE

HYDROLOGY, BIODIVERSITY AND STRUCTURAL CHANGE

MAP → recharge, passages, springs and values

AVOID → irreversible quarry and pollution pathways

MANAGE → visitation, waste and water use

MONITOR → hydrology, biodiversity and structural change

DEFINITION / COMPONENTS

- MAP → recharge, passages, springs and values
- AVOID → irreversible quarry and pollution pathways

TRIGGER / MECHANISM

- MANAGE → visitation, waste and water use
- MONITOR → hydrology, biodiversity and structural change

EFFECT / INTERVENTION

- MAP → recharge, passages, springs and values
- AVOID → irreversible quarry and pollution pathways

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MANAGE → visitation, waste and water use; MONITOR → hydrology, biodiversity and structural change.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

MEGHALAYA HAS INDIA'S PREMIER KARST SYSTEMS. KREM LIAT PRAH

THE CHERRAPUNJI LIMESTONE CAVE SHOWS WELL-DEVELOPED STALACTITE AND STALAGMITE

CAVE TOURISM SPANS TWO DIFFERENT CATEGORIES

NATURAL KARST CAVES IN MEGHALAYA AND ROCK-CUT/

VERY HIGH RAINFALL OVER THICK LIMESTONE MAKES MEGHALAYA IDEAL

THESE CAVES DOUBLE AS PALAEOCLIMATE ARCHIVES — THEIR STALAGMITES

GLOBAL BOUNDARY STRATOTYPE SECTION AND POINT — THE PHYSICAL

OPTIONAL PROCESS DEPTH

- Meghalaya has India's premier karst systems. Krem Liat Prah is widely cited as India's longest
- The Cherrapunji limestone cave shows well-developed stalactite and stalagmite formations (dripstone deposition).
- Cave tourism spans two different categories: natural karst caves in Meghalaya and rock-cut/
- Very high rainfall over thick limestone makes Meghalaya ideal for rapid solution and cave growth.

DATA / INSTITUTION LIMIT

- These caves double as palaeoclimate archives — their stalagmites record past rainfall/drought.
- GSSP = Global Boundary Stratotype Section and Point — the physical 'golden spike' defining a time boundary.
- The Meghalayan Age is the youngest stage of the Holocene (last ~4,200 years); its GSSP is the KNI-51 stalagmite in Mawmluh Cave, Meghalaya.
- The stalagmite isotope signal marks abrupt weakening of the Indian summer monsoon around 4.2 ka.

CONTEMPORARY PARALLEL

- Krem Liat Prah in Meghalaya is widely cited as India's longest natural cave; mapped length is survey-dependent.
- Cherrapunji limestone cave: developed stalactite/stalagmite (Khullar).
- Ajanta (Maharashtra), Pachmarhi (MP ~500 caves), Khandagiri/Udayagiri (Odisha).
- High rainfall + thick limestone = rapid karst solution in Meghalaya.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.